

# Rapport-preuve - KGE Models (provenance : évalué vs mentionné)

Corpus : **55 articles**. Évalués validés vs tables\_only, mentionnés vs no\_tables (prose). Matching déterministe.

## Scores

| Score                                  | Valeur       |
|----------------------------------------|--------------|
| Précision évalués                      | <b>94.4%</b> |
| Précision mentionnés                   | <b>97.9%</b> |
| Recall évalués (indicatif)             | <b>95.0%</b> |
| Recall mentionnés (indicatif)          | <b>97.5%</b> |
| Évalués vérifiés / mentionnés vérifiés | 340 / 235    |

## A. Preuve - modèles/méthodes ÉVALUÉS (340)

| Article         | Entité       | Trouv é | Extrait (tableau)                                                             |   |               |
|-----------------|--------------|---------|-------------------------------------------------------------------------------|---|---------------|
| AdaptativeNS.md | ComplEx      | oui     | tj\$</td> </tr> <tr> <td>ComplEx</td><br><td>\$Re(\mathbf{h}^T \text          |   |               |
| AdaptativeNS.md | ConvE        | oui     | ))\$</td> </tr> <tr> <td>ConvE</td><br><td>\$f(\text{vec}(\mathbf{r},         |   |               |
| AdaptativeNS.md | DistMult     | oui     | tj\$</td> </tr> <tr> <td>DistMult</td><br><td>\$\mathbf{h}^T \text{di         |   |               |
| AdaptativeNS.md | MuRP         | oui     | ))\$</td> </tr> <tr> <td>MuRp</td><br><td>\$-d_{\mathbb{B}}\{\exp_{\{         |   |               |
| AdaptativeNS.md | RotatE       | oui     | 2}\$</td> </tr> <tr> <td>RotatE</td><br><td>\$\                               | \ | \mathbf{h} \} |
| AdaptativeNS.md | TransE       | oui     | d>-</td> </tr> <tr> <td>transE</td><br><td>0.226</td> <td>                    |   |               |
| ADNS.md         | DistMult     | oui     | e 3. Evaluation of Negative Sampling<br>on DistMult <table> <thead> <tr> <th> |   |               |
| ADNS.md         | TransE       | oui     | 2.** Evaluation of Negative Sampling<br>on TransE <table> <thead> <tr> <th>   |   |               |
| ASA.md          | ComplEx      | oui     | </thead> <tbody> <tr><br><td>ComplEx</td><br><td>53.18(10)</td>               |   |               |
| ASA.md          | ConvE        | oui     | (8)</td> </tr> <tr> <td>ConvE</td><br><td>49.65(11)</td>                      |   |               |
| ASA.md          | DistMult     | oui     | 10)</td> </tr> <tr> <td>DistMult</td><br><td>53.94(9)</td>                    |   |               |
| ASA.md          | GATNE        | oui     | (9)</td> </tr> <tr> <td>GATNE</td><br><td><u>96.25</u>(5)</td>                |   |               |
| ASA.md          | HAN          | oui     | 11)</td> </tr> <tr> <td>HAN</td><br><td>87.57(7)</td>                         |   |               |
| ASA.md          | metapath2vec | oui     | (5)</td> </tr> <tr><br><td>metapath2vec</td><br><td><u>94.15</u>(6)</td>      |   |               |
| ASA.md          | R-GCN        | oui     | (6)</td> </tr> <tr> <td>R-GCN</td><br><td>97.16(4)</td>                       |   |               |
| ASA.md          | RelGNN       | oui     | (3)</td> </tr> <tr> <td>RelGNN -<br>ASA</td> <td>98.84(2)</td>                |   |               |

| Article    | Entité                   | Trouvé | Extrait (tableau)                                                                    |  |  |
|------------|--------------------------|--------|--------------------------------------------------------------------------------------|--|--|
| BatchNS.md | ComplEx                  | oui    | gt;</td> </tr> <tr> <td>ComplEx</td><br><td>x ■ θ■</td> <t                           |  |  |
| BatchNS.md | DeepWalk                 | oui    | </thead> <tbody> <tr><br><td>DeepWalk*</td> <td>0.691</td><br><t                     |  |  |
| BatchNS.md | HolE                     | oui    | 749</td> </tr> <tr> <td>HolE (Nickel<br>et al., 2016b)</td> <td>                     |  |  |
| BatchNS.md | MILE (1 level)           | oui    | GB</td> </tr> <tr> <td>MILE (1<br>level)*</td> <td>0.629</td> <                      |  |  |
| BatchNS.md | MILE (5 levels)          | oui    | GB</td> </tr> <tr> <td>MILE (5<br>levels)*</td> <td>0.505</td> <                     |  |  |
| BatchNS.md | PBG                      | oui    | GB</td> </tr> <tr> <td>PBG (1<br>partition)</td> <td>0.749</t                        |  |  |
| BatchNS.md | PBG (ComplEx)            | oui    | 785</td> </tr> <tr> <td>PBG<br>(ComplEx)</td> <td>0.242</td> <t                      |  |  |
| BatchNS.md | PBG (TransE)             | oui    | 910</td> </tr> <tr> <td>PBG<br>(TransE)</td> <td>0.265</td> <t                       |  |  |
| BatchNS.md | R-GCN+                   | oui    | 840</td> </tr> <tr> <td>R-GCN+<br>(Schlichtkrull et al., 2018)</td>                  |  |  |
| BatchNS.md | Reciprocal<br>ComplEx-N3 | oui    | 838</td> </tr> <tr> <td>Reciprocal<br>ComplEx-N3 (Lacroix et al.,<br>2018)</td> <td> |  |  |
| BatchNS.md | RESCAL                   | oui    | </thead> <tbody> <tr><br><td>RESCAL</td> <td>A■x</td><br><td>&                       |  |  |
| BatchNS.md | StarSpace                | oui    | 842</td> </tr> <tr> <td>StarSpace<br>(Wu et al., 2018)</td> <td>.</t                 |  |  |
| BatchNS.md | TransE                   | oui    | gt;</td> </tr> <tr> <td>TransE</td><br><td>x + θ■</td> <t                            |  |  |
| CAKE.md    | HAKE                     | oui    | ng></td> </tr> <tr> <td>HAKE</td><br><td>34</td> <td>0.                              |  |  |
| CAKE.md    | RotatE                   | oui    | ng></td> </tr> <tr> <td>RotatE</td><br><td>35</td> <td>0.                            |  |  |
| CAKE.md    | TransE                   | oui    | </thead> <tbody> <tr><br><td>TransE</td> <td>35</td> <td>0.                          |  |  |
| CANS.md    | CKRL                     | oui    | 9.9</td> </tr> <tr><br><td>CKRL(LT)</td> <td>230</td> <                              |  |  |
| CANS.md    | TransE                   | oui    | </thead> <tbody> <tr><br><td>TransE</td> <td>249</td> <td>1                          |  |  |
| CCS.md     | Op-TransD                | oui    | 873</td> </tr> <tr><br><td>Op-TransD</td> <td>3352</td><br><td>                      |  |  |
| CCS.md     | Op-TransE                | oui    | </thead> <tbody> <tr><br><td>Op-TransE</td> <td>4171</td><br><td>                    |  |  |
| CCS.md     | Op-TransH                | oui    | 852</td> </tr> <tr><br><td>Op-TransH</td> <td>4404</td><br><td>                      |  |  |
| CCS.md     | Op-TransR                | oui    | 815</td> </tr> <tr><br><td>Op-TransR</td> <td>3685</td><br><td>                      |  |  |
| CCS.md     | TransD                   | oui    | g functions for TransE, TransH,<br>TransR, TransD. <table> <thead><br><tr> <th       |  |  |

| Article            | Entité   | Trouv é | Extrait (tableau)                                                                        |  |  |
|--------------------|----------|---------|------------------------------------------------------------------------------------------|--|--|
| CCS.md             | TransE   | oui     | 4. Definition of scoring functions for TransE, TransH, TransR, TransD. <table> <thead>   |  |  |
| CCS.md             | TransH   | oui     | inition of scoring functions for TransE, TransH, TransR, TransD. <table> <thead>         |  |  |
| CCS.md             | TransR   | oui     | f scoring functions for TransE, TransH, TransR, TransD. <table> <thead> <tr>             |  |  |
| ConceptDriven.md   | DistMult | oui     | >81.4</td> <td>4.15</td> </tr> <tr> <td>DistMult [18]</td> <td>88.4</td> <td>86.1</td> < |  |  |
| ConceptDriven.md   | DKRL     | oui     | d>44.5</td> <td>52.2</td> </tr><tr> <td>DKRL [16]</td> <td>26.0</td> <td>100.00</td>     |  |  |
| ConceptDriven.md   | RotatE   | oui     | d>43.2</td> <td>50.7</td> </tr><tr> <td>RotatE [17]</td> <td>47.6</td> <td>100.00</td>   |  |  |
| ConceptDriven.md   | TransE   | oui     | 0↑</th> </tr> </thead> <tbody> <tr> <td>TransE [3]</td> <td>24.3</td> <td>100.00</td>    |  |  |
| ConceptDriven.md   | TransH   | oui     | d>40.6</td> <td>47.6</td> </tr><tr> <td>TransH [14]</td> <td>23.1</td> <td>100.00</td>   |  |  |
| ConceptDriven.md   | TransR   | oui     | d>44.5</td> <td>52.2</td> </tr><tr> <td>TransR [15]</td> <td>23.2</td> <td>100.00</td>   |  |  |
| CondConstraints.md | TransE   | NON     | -                                                                                        |  |  |
| dans.md            | ComplEx  | oui     | RGCN) and decoders (DistMult, RotatE or ComplEx). The best results are in bold, and the  |  |  |
| dans.md            | DistMult | oui     | the same backbone (RGCN) and decoders (DistMult, RotatE or ComplEx). The best results a  |  |  |
| dans.md            | RotatE   | oui     | backbone (RGCN) and decoders (DistMult, RotatE or ComplEx). The best results are in bo   |  |  |
| DeMix.md           | ComplEx  | oui     | ng></td> </tr> <tr> <td>ComplEx+Uniform</td> <td>0.429</td>                              |  |  |
| DeMix.md           | DistMult | oui     | </thead> <tbody> <tr> <td>DistMult+Uniform</td> <td>0.412</td>                           |  |  |
| DeMix.md           | HAKE     | oui     | ng></td> </tr> <tr> <td>HAKE+Unifor m<sup> $\Delta$ </td> <td>0.                         |  |  |
| DeMix.md           | RotatE   | oui     | </thead> <tbody> <tr> <td>RotatE+U niform<sup> $\Delta$ </td> <td>0.                     |  |  |
| DeMix.md           | TransE   | oui     | les</td> </tr> <tr> <td>TransE+Uniform</td> <td>0.175*</td>                              |  |  |
| DHNS.md            | AdaMF    | oui     | .88</td> </tr> <tr> <td>AdaMF</td> <td><u>32.51</u></td>                                 |  |  |
| DHNS.md            | ComplEx  | oui     | .80</td> </tr> <tr> <td>ComplEx</td> <td>27.48</td> <td>                                 |  |  |
| DHNS.md            | DistMult | oui     | .68</td> </tr> <tr> <td>DistMult</td> <td>23.03</td> <td>                                |  |  |

| Article     | Entité   | Trouvé | Extrait (tableau)                                                                            |  |  |
|-------------|----------|--------|----------------------------------------------------------------------------------------------|--|--|
| DHNS.md     | GC-OTE   | oui    | .71</td> </tr> <tr> <td>GC-OTE</td><br><td>31.85</td> <td>                                   |  |  |
| DHNS.md     | IKRL     | oui    | wspace="8">MMKGC Models</td><br><td>IKRL</td> <td>26.82</td> <td>                            |  |  |
| DHNS.md     | MMKRL    | oui    | .72</td> </tr> <tr> <td>MMKRL</td><br><td>26.81</td> <td>                                    |  |  |
| DHNS.md     | OTKGE    | oui    | .62</td> </tr> <tr> <td>OTKGE</td><br><td>23.86</td> <td>                                    |  |  |
| DHNS.md     | PairRE   | oui    | .73</td> </tr> <tr> <td>PairRE</td><br><td>31.13</td> <td>                                   |  |  |
| DHNS.md     | RotatE   | oui    | r framework DHNS</mark> integrated<br>with RotatE model and some<br>state-of-the-art baselin |  |  |
| DHNS.md     | RSME     | oui    | .44</td> </tr> <tr> <td>RSME</td><br><td>29.76</td> <td>                                     |  |  |
| DHNS.md     | TBKGC    | oui    | .07</td> </tr> <tr> <td>TBKGC</td><br><td>28.08</td> <td>                                    |  |  |
| DHNS.md     | TransAE  | oui    | .58</td> </tr> <tr> <td>TransAE</td><br><td>28.09</td> <td>                                  |  |  |
| DHNS.md     | TransD   | oui    | .23</td> </tr> <tr> <td>TransD</td><br><td>21.52</td> <td>                                   |  |  |
| DHNS.md     | TransE   | oui    | 7">Unimodal KGC Models</td><br><td>TransE</td> <td>24.86</td> <td>                           |  |  |
| DHNS.md     | VBKGC    | oui    | .83</td> </tr> <tr> <td>VBKGC</td><br><td>30.61</td> <td>                                    |  |  |
| DMNS.md     | ARVGA    | oui    | 005</td> </tr> <tr> <td>ARVGA</td><br><td>.732 ± .011</td>                                   |  |  |
| DMNS.md     | GAT      | oui    | </thead> <tbody> <tr> <td>GAT</td><br><td>.766 ± .006</td>                                   |  |  |
| DMNS.md     | GCN      | oui    | link prediction against baselines using<br>GCN as the base encoder. <table><br><thead>       |  |  |
| DMNS.md     | GraphGAN | oui    | 004</td> </tr> <tr><br><td>GraphGAN</td> <td>.739 ±<br>.003</td>                             |  |  |
| DMNS.md     | GVAE     | oui    | 003</td> </tr> <tr> <td>GVAE</td><br><td><u>.783 ± .003</u></td>                             |  |  |
| DMNS.md     | SAGE     | oui    | ng></td> </tr> <tr> <td>SAGE</td><br><td>.598 ± .014</td>                                    |  |  |
| DMNS.md     | ScaLed   | oui    | 001</td> </tr> <tr> <td>ScaLed</td><br><td>.676 ± .004</td>                                  |  |  |
| DMNS.md     | SEAL     | oui    | 002</td> </tr> <tr> <td>SEAL</td><br><td>.751 ± .007</td>                                    |  |  |
| DNS.md      | RESCAL   | oui    | parameters. RNS results for TransE<br>and RESCAL are borrowed from<br>Nickel et al. [2016b]  |  |  |
| DNS.md      | TransE   | oui    | its default parameters. RNS results<br>for TransE and RESCAL are<br>borrowed from Nickel et  |  |  |
| DomainNS.md | CTransR  | oui    | 8.7</td> </tr> <tr> <td>CTransR (Lin<br>et al., 2015b)</td> <td>No                           |  |  |
| DomainNS.md | DISTMULT | oui    | 1.4</td> </tr> <tr> <td>DISTMULT<br>(Yang et al., 2015)</td> <td>No                          |  |  |
| DomainNS.md | IRN      | oui    | 4.7</td> </tr> <tr> <td>IRN (Shen et<br>al., 2016)</td> <td>Ex                               |  |  |

| Article     | Entité                | Trouvé | Extrait (tableau)                                                                          |  |  |
|-------------|-----------------------|--------|--------------------------------------------------------------------------------------------|--|--|
| DomainNS.md | ITransF               | oui    | </tr> <tr><br><td><strong>ITransF</strong></td><br><td>No</td>                             |  |  |
| DomainNS.md | NLFeat                | oui    | 4.6</td> </tr> <tr> <td>NLFeat<br>(Toutanova and Chen, 2015)</td>                          |  |  |
| DomainNS.md | PTransE               | oui    | 6.2</td> </tr> <tr> <td>PTransE (Lin<br>et al., 2015a)</td> <td>Pa                         |  |  |
| DomainNS.md | Random Walk           | oui    | 7.0</td> </tr> <tr> <td>Random Walk<br>(Wei et al., 2016)</td> <td>Pat                     |  |  |
| DomainNS.md | RTransE               | oui    | ng></td> </tr> <tr> <td>RTransE<br>(García-Durán et al., 2015)</td>                        |  |  |
| DomainNS.md | STransE               | oui    | the brackets are another set of results<br>STransE reported. <table> <thead><br><tr>       |  |  |
| DomainNS.md | TATEC                 | oui    | 7.3</td> </tr> <tr> <td>TATEC<br>(García-Durán et al., 2016)</td>                          |  |  |
| DomainNS.md | TransD                | oui    | 4.0</td> </tr> <tr> <td>TransD (Ji et<br>al., 2015)</td> <td>No</                          |  |  |
| DomainNS.md | TransE                | oui    | 6.3</td> </tr> <tr> <td>TransE<br>(Bordes et al., 2013)</td> <td>                          |  |  |
| DomainNS.md | TransH                | oui    | 7.1</td> </tr> <tr> <td>TransH (Wang<br>et al., 2014)</td> <td>No                          |  |  |
| DomainNS.md | TransR                | oui    | 4.4</td> </tr> <tr> <td>TransR (Lin et<br>al., 2015b)</td> <td>No                          |  |  |
| EANS.md     | ComplEx               | oui    | </tr> <tr> <td<br>rowspan="6">ComplEx<br/>(Trouillon<br>et al., 2016)</td>                 |  |  |
| EANS.md     | DistMult              | oui    | </tr> <tr> <td<br>rowspan="7">DistMult<br/>(Yang et<br>al., 2014)</td> <t                  |  |  |
| EANS.md     | RotatE                | oui    | </tr> <tr> <td<br>rowspan="5">RotatE<br/>(Sun et al.,<br>2019)</td> <td                    |  |  |
| EANS.md     | TransD                | oui    | </tr> <tr> <td<br>rowspan="6">TransD<br/>(Ji et al.,<br>2015)</td> <td>                    |  |  |
| EANS.md     | TransE                | oui    | tbody> <tr> <td<br>rowspan="7">TransE<br/>(Bordes et<br>al., 2013)</td>                    |  |  |
| ERDNS.md    | CompGCN<br>(ConvE)    | oui    | </tr> <tr> <td<br>rowspan="3">CompGCN<br>(ConvE)</td> <td>unif</td> <td                    |  |  |
| ERDNS.md    | CompGCN<br>(DistMult) | oui    | </tr> <tr> <td<br>rowspan="3">CompGCN<br>(DistMult)</td> <td>unif</td> <td                 |  |  |
| ERDNS.md    | CompGCN<br>(TransE)   | oui    | </tr> <tr> <td<br>rowspan="3">CompGCN<br>(TransE)</td> <td>unif</td> <td                   |  |  |
| ERDNS.md    | ComplEx               | oui    | 3. Performance comparison on<br>DistMult, ComplEx, TransE and<br>RotatE. <table> <thead>   |  |  |
| ERDNS.md    | ConvE                 | oui    | le> Table 4. Performance comparison<br>on ConvE and CompGCN. <table><br><thead> <tr>       |  |  |
| ERDNS.md    | DistMult              | oui    | le> Table 3. Performance comparison<br>on DistMult, ComplEx, TransE and<br>RotatE. <table> |  |  |

| Article          | Entité      | Trouvé     | Extrait (tableau)                                                               |  |  |
|------------------|-------------|------------|---------------------------------------------------------------------------------|--|--|
| ERDNS.md         | RotatE      | oui        | arison on DistMult, ComplEx, TransE and RotatE. <table> <thead> <tr> <th>       |  |  |
| ERDNS.md         | TransE      | oui        | rmance comparison on DistMult, ComplEx, TransE and RotatE. <table> <thead> <tr> |  |  |
| eTruncatedUNS.md | AlignE      | oui        | 320</td> </tr> <tr> <td>AlignE</td> <td>47.18</td> <td>                         |  |  |
| eTruncatedUNS.md | BootEA      | oui        | 707</td> </tr> <tr> <td>BootEA</td> <td><strong>62.94</strong>                  |  |  |
| eTruncatedUNS.md | IPTransE    | oui        | 334</td> </tr> <tr> <td>IPTransE</td> <td>40.59</td> <td>                       |  |  |
| eTruncatedUNS.md | JAPE        | oui        | 386</td> </tr> <tr> <td>JAPE</td> <td>41.18</td> <td>                           |  |  |
| eTruncatedUNS.md | MTransE     | oui        | </thead> <tbody> <tr> <td>MTransE</td> <td>30.83</td> <td>                      |  |  |
| GHN.md           | DistMult    | oui        | 4.1</td> </tr> <tr> <td>DistMult</td> <td>44.4</td> <td>                        |  |  |
| GHN.md           | DKRL        | oui        | em></td> </tr> <tr> <td>DKRL</td> <td>16.0</td> <td>                            |  |  |
| GHN.md           | KEPLER      | oui        | es\$</td> </tr> <tr> <td>KEPLER</td> <td>21.0</td> <td>                         |  |  |
| GHN.md           | KG-BERT     | oui        | ods</td> </tr> <tr> <td>KG-BERT</td> <td>21.6</td> <td>                         |  |  |
| GHN.md           | MTL-KGC     | oui        | 2.0</td> </tr> <tr> <td>MTL-KGC</td> <td>33.1</td> <td>                         |  |  |
| GHN.md           | RotatE      | oui        | 4.6</td> </tr> <tr> <td>RotatE</td> <td>47.6</td> <td>                          |  |  |
| GHN.md           | SimKGC      | oui        | 8.2</td> </tr> <tr> <td>SimKGC</td> <td>66.6</td> <td>                          |  |  |
| GHN.md           | StAR        | oui        | 5.8</td> </tr> <tr> <td>StAR</td> <td>40.1</td> <td>                            |  |  |
| GHN.md           | TransE      | oui        | ods</td> </tr> <tr> <td>TransE</td> <td>24.3</td> <td>                          |  |  |
| GHN.md           | Tucker      | oui        | 3.3</td> </tr> <tr> <td>TuckER</td> <td>47.0</td> <td>                          |  |  |
| GibbsNS.md       | ComplEx     | oui        | 533</td> </tr> <tr> <td>ComplEx</td> <td>0.323</td> <td>                        |  |  |
| GibbsNS.md       | ConvE       | oui        | 526</td> </tr> <tr> <td>ConvE</td> <td>0.325</td> <td>                          |  |  |
| GibbsNS.md       | DistMult    | oui        | d>.</td> </tr> <tr> <td>DistMult</td> <td>0.308</td> <td>                       |  |  |
| GibbsNS.md       | InteractE   | oui        | .52</td> </tr> <tr> <td>InteractE</td> <td>0.354</td> <td>                      |  |  |
| GibbsNS.md       | KGE-EML     | oui        | 569</td> </tr> <tr> <td>KGE-EML [11]</td> <td>0.36</td> <td>                    |  |  |
| GibbsNS.md       | KGE-EML+GNS | <b>NON</b> | -                                                                               |  |  |
| GibbsNS.md       | PairRE      | oui        | ng></td> </tr> <tr> <td>PairRE</td> <td>0.351</td> <td>                         |  |  |
| GibbsNS.md       | PROCRUSTES  | oui        | 528</td> </tr> <tr> <td>PROCRUSTES</td> <td>0.345</td> <td>                     |  |  |

| Article     | Entité                      | Trouv é    | Extrait (tableau)                                                         |  |  |
|-------------|-----------------------------|------------|---------------------------------------------------------------------------|--|--|
| GibbsNS.md  | RotatE                      | oui        | 529</td> </tr> <tr> <td>RotatE</td><br><td>0.338</td> <td>                |  |  |
| GibbsNS.md  | TransD                      | oui        | </tr> <tr> <td<br>rowspan="5">TransD</td><br><td>Uniform</td> <           |  |  |
| GibbsNS.md  | TransE                      | oui        | </thead> <tbody> <tr><br><td>TransE</td> <td>0.33</td> <td>               |  |  |
| GibbsNS.md  | TuckER                      | oui        | 554</td> </tr> <tr> <td>TuckER</td><br><td>0.358</td> <td>                |  |  |
| GNDN.md     | ConvE                       | oui        | 5.8</td> </tr> <tr> <td>ConvE<br>[19]*</td> <td>0.46</td>                 |  |  |
| GNDN.md     | ConvKB                      | oui        | 9.1</td> </tr> <tr> <td>ConvKB<br>[20]</td> <td>0.248</td>                |  |  |
| GNDN.md     | Knowledge<br>Completion GAN | <b>NON</b> | -                                                                         |  |  |
| GNDN.md     | NTN                         | oui        | 5.2</td> </tr> <tr> <td>NTN [8]*</td><br><td>70.6</td>                    |  |  |
| GNDN.md     | Path-KCGAN [2<br>hop]       | oui        | 4.6</td> </tr> <tr><br><td>Path-KCGAN[2 hop]</td><br><td>0.58</td> <td>   |  |  |
| GNDN.md     | Path-KCGAN [3<br>hop]       | oui        | 7.1</td> </tr> <tr><br><td>Path-KCGAN[3 hop]</td><br><td>0.60</td> <td>   |  |  |
| GNDN.md     | PathG-RNN [2<br>hop]        | oui        | </thead> <tbody> <tr><br><td>PathG-RNN [2 hop]</td><br><td>0.41</td> <td> |  |  |
| GNDN.md     | PathG-RNN [3<br>hop]        | oui        | 6.3</td> </tr> <tr> <td>PathG-RNN [3<br>hop]</td> <td>0.43</td> <td>      |  |  |
| GNDN.md     | PTransE [2 hop]             | oui        | 8.5</td> </tr> <tr> <td>PTransE [2<br>hop]</td> <td>0.49</td> <td>        |  |  |
| GNDN.md     | PTransE [3 hop]             | oui        | 2.2</td> </tr> <tr> <td>PTransE [3<br>hop]</td> <td>0.54</td> <td>        |  |  |
| GNDN.md     | TransE                      | oui        | </thead> <tbody> <tr> <td>TransE<br>[9]*</td> <td>70.9</td>               |  |  |
| GNDN.md     | TransG                      | oui        | 8.9</td> </tr> <tr> <td>TransG<br>[16]*</td> <td>87.4</td>                |  |  |
| GNS.md      | KG-BERT                     | <b>NON</b> | -                                                                         |  |  |
| GraphGAN.md | DeepWalk                    | oui        | </thead> <tbody> <tr><br><td>DeepWalk</td> <td>0.841</td><br><td>         |  |  |
| GraphGAN.md | GraphGAN                    | oui        | 776</td> </tr> <tr><br><td>GraphGAN</td><br><td><strong>0.855</strong>    |  |  |
| GraphGAN.md | LINE                        | oui        | 812</td> </tr> <tr> <td>LINE</td><br><td>0.820</td> <td>                  |  |  |
| GraphGAN.md | Node2vec                    | oui        | 761</td> </tr> <tr><br><td>Node2vec</td> <td>0.845</td><br><td>           |  |  |
| GraphGAN.md | Struc2vec                   | oui        | 842</td> </tr> <tr><br><td>Struc2vec</td> <td>0.821</td><br><td>          |  |  |
| HaSa.md     | ComplEx                     | <b>NON</b> | -                                                                         |  |  |
| HaSa.md     | DistMult                    | oui        | ng></td> </tr> <tr> <td>DistMult<br>[36]</td> <td>7,000</td>              |  |  |

| Article  | Entité               | Trouvé | Extrait (tableau)                                                                                   |  |  |
|----------|----------------------|--------|-----------------------------------------------------------------------------------------------------|--|--|
| HaSa.md  | HaSa+                | oui    | /tr> <tr> <td>\$\mathcal{L}_{\{HaSa\}}\$<br>(6)</td> <td><u>123</u></td>                            |  |  |
| HaSa.md  | KG-BERT              | oui    | KGE</td> </tr> <tr> <td>KGBERT<br>[38]</td> <td>97</td> <                                           |  |  |
| HaSa.md  | LASS                 | oui    | 482</td> </tr> <tr> <td>LASS<br>[25]</td> <td>55</td> <                                             |  |  |
| HaSa.md  | RotatE               | oui    | 441</td> </tr> <tr> <td>RotatE<br>[26]</td> <td>3,340</td>                                          |  |  |
| HaSa.md  | sentence-BERT        | oui    | 6\$). All methods use the pre-trained<br>LM sentence-BERT as the<br>initialization. Boldface is the |  |  |
| HaSa.md  | StAR                 | oui    | 420</td> </tr> <tr> <td>StAR<br>[30]</td> <td><strong>51</strong>                                   |  |  |
| HaSa.md  | TransE               | oui    | KGE</td> </tr> <tr> <td>TransE<br>[3]</td> <td>2,300</td>                                           |  |  |
| HTENS.md | ComplEx              | oui    | )d\$</td> </tr> <tr> <td>ComplEx</td><br><td>\$\mathbf{h}, \mathbf{m}                               |  |  |
| HTENS.md | DistMult             | oui    | )d\$</td> </tr> <tr> <td>DistMult</td><br><td>\$\mathbf{h}, \mathbf{m}                              |  |  |
| HTENS.md | TransD               | oui    | )d\$</td> </tr> <tr> <td>TransD</td><br><td>\$\mathbf{l} + \mathbf{m}                               |  |  |
| HTENS.md | TransE               | oui    | </thead> <tbody> <tr><br><td>TransE</td> <td>\$\mathbf{h} + \mathbf{m}                              |  |  |
| IGAN.md  | Single Layer Model   | NON    | -                                                                                                   |  |  |
| IGAN.md  | SME(Bilinear)        | oui    | ng></td> </tr> <tr> <td>SME(Bilinear)<br>(Bordes et al. 2012)</td> <td>                             |  |  |
| IGAN.md  | Structured Embedding | NON    | -                                                                                                   |  |  |
| IGAN.md  | TransD               | oui    | ng></td> </tr> <tr> <td>TransD (Ji et<br>al. 2015)</td> <td>91</td>                                 |  |  |
| IGAN.md  | TransE               | oui    | 5.7</td> </tr> <tr> <td>TransE<br>(Bordes et al. 2013)</td> <td><                                   |  |  |
| IGAN.md  | TransE + A-LSTM      | NON    | -                                                                                                   |  |  |
| IGAN.md  | TransH               | oui    | ng></td> </tr> <tr> <td>TransH<br>(Wang et al. 2014)</td> <td>87<                                   |  |  |
| IGAN.md  | TransR               | oui    | ng></td> </tr> <tr> <td>TransR (Lin<br>et al. 2015)</td> <td><str                                   |  |  |
| IGAN.md  | Unstructured Model   | NON    | -                                                                                                   |  |  |
| KBGAN.md | ComplEx              | oui    | re considered in this paper. Except for<br>COMPLEX, all boldface lower case<br>letters repres       |  |  |
| KBGAN.md | DistMult             | oui    | }\mathbf{l} + \mathbf{m}                                                                            |  |  |
| KBGAN.md | TransD               | oui    | }\mathbf{l} + \mathbf{m}                                                                            |  |  |
| KBGAN.md | TransE               | oui    | </thead> <tbody> <tr><br><td>TRANSE</td> <td>\$\mathbf{h} + \mathbf{m}                              |  |  |
| KSGAN.md | ComplEx              | oui    | ^d\$</td> </tr> <tr> <td>ComplEx</td><br><td>\$\mathbf{h}, \mathbf{m}                               |  |  |



| Article           | Entité     | Trouvé | Extrait (tableau)                                                                                 |                             |  |
|-------------------|------------|--------|---------------------------------------------------------------------------------------------------|-----------------------------|--|
| KSGAN.md          | TransD     | oui    | d]\$/td> </tr> <tr> <td>TransD</td><br><td>\$-\                                                   | (w_r w_h^\top + I)          |  |
| KSGAN.md          | TransE     | oui    | br/>distance<br/>model</td><br><td>TransE</td> <td>\$-\                                           | \mathbf{h} + \mathbf{h}     |  |
| LAS.md            | ComplEx    | oui    | 462</td> </tr> <tr> <td>ComplEx<br>(pretrained)</td> <td>0.264</td>                               |                             |  |
| LAS.md            | ConvE      | oui    | d>.</td> </tr> <tr> <td>ConvE<br>[23]</td> <td><strong>31.6</str                                  |                             |  |
| LAS.md            | DistMult   | oui    | </thead> <tbody> <tr> <td>DistMult<br>[18]</td> <td>24.1</td>                                     |                             |  |
| LAS.md            | R-GCN      | oui    | 1.0</td> </tr> <tr> <td>R-GCN<br>[28]</td> <td>24.8</td>                                          |                             |  |
| LAS.md            | TransD*    | oui    | 431</td> </tr> <tr> <td>TransD<br>(pretrained)</td> <td>0.244</td>                                |                             |  |
| LAS.md            | TransE     | oui    | </thead> <tbody> <tr> <td>TransE<br>(pretrained)</td> <td>0.243</td>                              |                             |  |
| LEMON.md          | RotatE     | oui    | tbody> <tr> <td<br>rowspan="4">RotatE</td><br><td>Uniform</td> <                                  |                             |  |
| LEMON.md          | TransE     | oui    | </tr> <tr> <td<br>rowspan="4">TransE</td><br><td>Uniform</td> <                                   |                             |  |
| Localcognitive.md | ComplEx    | oui    | \$\circ\$ denotes element-wise<br>(Hadamard) complex product,<br>\$\otimes\$ denotes element-wise |                             |  |
| Localcognitive.md | ConvE      | oui    | 599</td> </tr> <tr> <td>ConvE</td><br><td>374</td> <td>.                                          |                             |  |
| Localcognitive.md | DistMult   | oui    | ="3">Semantic matching</td><br><td>DistMult</td> <td>\$\angle\$<br>\mathbf{r}, \mathbf{h}         |                             |  |
| Localcognitive.md | HolE       | oui    | d>.</td> </tr> <tr> <td>HolE</td><br><td>.</td> <td>.93                                           |                             |  |
| Localcognitive.md | QuatE      | oui    | d>.</td> </tr> <tr> <td>QuatE</td><br><td>\$\mathbf{h} \otimes \mathbf{h}                         |                             |  |
| Localcognitive.md | RatE       | oui    | d>\$\times\$</td> </tr> <tr> <td>RatE</td><br><td>\$-\                                            | \mathbf{h} \odot \mathbf{h} |  |
| Localcognitive.md | RotatE     | oui    | d>\$\times\$</td> </tr> <tr> <td>RotatE</td><br><td>\$-\                                          | \mathbf{h} \circ \mathbf{h} |  |
| Localcognitive.md | TransE     | oui    | owspan="3">Trans-based</td><br><td>TransE</td> <td>\$-\                                           | \mathbf{h} + \mathbf{h}     |  |
| Localcognitive.md | TuckER     | oui    | Table 9: Performance comparison<br>between TuckER and RatE on<br>WN18RR/FB15k-237. “#0” deno      |                             |  |
| LTS.md            | CTransR    | oui    | 8.7</td> </tr> <tr><br><td>CTransR(unif/bern)</td><br><td>243/231</td>                            |                             |  |
| LTS.md            | SE         | oui    | ilt</th> </tr> <tr> <td>SE</td><br><td>1011</td> <td>                                             |                             |  |
| LTS.md            | TransD     | oui    | 0.2</td> </tr> <tr><br><td>TransD(unif/bern)</td><br><td>242/224</td>                             |                             |  |
| LTS.md            | TransE     | oui    | </thead> <tbody> <tr><br><td>TransE</td> <td>7.01</td> <td>                                       |                             |  |
| LTS.md            | TransE-LTS | oui    | .71</td> </tr> <tr><br><td>TransE-LTS</td> <td>13.89</td><br><td>                                 |                             |  |

| Article       | Entité     | Trouvé | Extrait (tableau)                                                                                             |                                                                       |                                        |
|---------------|------------|--------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------|
| LTS.md        | TransE-SNS | oui    | >74</td> </tr> <tr><br><td>TransE-SNS(unif/bern)</td><br><td>220/207</td>                                     |                                                                       |                                        |
| LTS.md        | TransH     | oui    | 7.1</td> </tr> <tr><br><td>TransH(unif/bern)</td><br><td>318/401</td>                                         |                                                                       |                                        |
| LTS.md        | TranSparse | oui    | 7.3</td> </tr> <tr><br><td>TranSparse(unif/bern)</td><br><td>233/223</td>                                     |                                                                       |                                        |
| LTS.md        | TransR     | oui    | 4.4</td> </tr> <tr><br><td>TransR(unif/bern)</td><br><td>232/238</td>                                         |                                                                       |                                        |
| M2ixKG.md     | ComplEx    | oui    | ct, $\bar{\cdot}$ denotes conjugate<br>for complex vectors <table> <thead><br><tr>                            |                                                                       |                                        |
| M2ixKG.md     | DistMult   | oui    | = "2">semantic matching</td><br><td>DistMult</td> <td> $\angle$<br>$\mathbf{r}$ , $\backslash$                |                                                                       |                                        |
| M2ixKG.md     | RotatE     | oui    | $\backslash$                                                                                                  | $\backslash$ </td> </tr> <tr><br><td>RotatE</td><br><td> $\backslash$ | $\mathbf{h}$<br>} $\odot$ $\mathbf{m}$ |
| M2ixKG.md     | TransE     | oui    | translational distance</td><br><td>TransE</td> <td> $\backslash$                                              | $\mathbf{h} + \mathbf{b}$                                             |                                        |
| MCNS.md       | DeepWalk   | oui    | <tr> <th> </th> <th>DeepWalk</th><br><th>GCN</th> <th>G                                                       |                                                                       |                                        |
| MCNS.md       | GCN        | oui    | > <th>DeepWalk</th> <th>GCN</th><br><th>GraphSAGE</th>                                                        |                                                                       |                                        |
| MCNS.md       | GraphSAGE  | oui    | k</th> <th>GCN</th><br><th>GraphSAGE</th><br><th>DeepWalk</th>                                                |                                                                       |                                        |
| MDNCaching.md | ComplEx    | oui    | $\wedge$ </td> </tr> <tr> <td>ComplEx<br>[11]</td> <td> $\text{Re}(\mathbf{h} \odot \text{diag}(\mathbf{r}))$ |                                                                       |                                        |
| MDNCaching.md | DistMult   | oui    | tic<br>matching-based</td><br><td>DistMult [16]</td> <td> $\mathbf{h} \odot \text{diag}(\mathbf{r})$          |                                                                       |                                        |
| MDNCaching.md | TransD     | oui    | $\wedge$ </td> </tr> <tr> <td>TransD<br>[6]</td> <td> $\mathbf{h} \odot \mathbf{r}$                           | $\mathbf{h} + \mathbf{w}_r \mathbf{w}_u$                              |                                        |
| MDNCaching.md | TransE     | oui    | nal<br>distance-based</td><br><td>TransE [2]</td> <td> $\mathbf{h} + \mathbf{r} - \mathbf{t}$                 | $\mathbf{h} + \mathbf{r} - \mathbf{t}$                                | $\mathbf{u}$                           |
| NMiss.md      | DistMult   | oui    | -06</td> </tr> <tr> <td>DistMult</td><br><td>0.001</td> <td>                                                  |                                                                       |                                        |
| NMiss.md      | RESICAL    | oui    | -06</td> </tr> <tr> <td>Rescal</td><br><td>0.001</td> <td>                                                    |                                                                       |                                        |
| NMiss.md      | TransE     | oui    | 084</td> </tr> <tr> <td>TransE</td><br><td>0.001</td> <td>                                                    |                                                                       |                                        |
| NoiGAN.md     | Attention  | oui    | 507</td> </tr> <tr> <td>Attention</td><br><td><strong>436</strong></td>                                       |                                                                       |                                        |
| NoiGAN.md     | DistMult   | oui    | <td rowspan="10">0%</td><br><td>DistMult</td> <td>.218</td> <td>                                              |                                                                       |                                        |
| NoiGAN.md     | RotatE     | oui    | 503</td> </tr> <tr> <td>RotatE</td><br><td>.326</td> <td>                                                     |                                                                       |                                        |
| NoiGAN.md     | TransE     | oui    | > </tr> <tr> <td>NoiGAN-TransE<br>(soft)</td> <td><strong>.949</strong></td>                                  |                                                                       |                                        |
| NSCaching.md  | ComplEx    | oui    | op</td> </tr> <tr> <td>ComplEx<br>[38]</td> <td> $\text{Re}(\mathbf{h} \odot \mathbf{r})$                     |                                                                       |                                        |

| Article          | Entité   | Trouv é | Extrait (tableau)                                                                            |                           |   |
|------------------|----------|---------|----------------------------------------------------------------------------------------------|---------------------------|---|
| NSCaching.md     | DistMult | oui     | = <sup>2</sup> semantic matching                                                             |                           |   |
| NSCaching.md     | TransD   | oui     | DistMult [46]<br>$\mathbf{h} \cdot \mathbf{h}$                                               | $\mathbf{h} + \mathbf{h}$ |   |
| NSCaching.md     | TransE   | oui     | translational distance<br>TransE [7]<br>$\mathbf{h} + \mathbf{m}$                            | $\mathbf{h} + \mathbf{m}$ |   |
| NSCaching.md     | TransH   | oui     | $\mathbf{h}$<br>TransH [42]<br>$\mathbf{h} - \mathbf{h}$                                     | $\mathbf{h} - \mathbf{h}$ |   |
| PNS.md           | TransE   | oui     | TransE                                                                                       | 263                       | 0 |
| PNS.md           | TransH   | oui     | TransH(bern)                                                                                 | 401                       |   |
| PNS.md           | TransR   | oui     | Epoch<br>Modified TransR<br>Original TransR                                                  |                           |   |
| RAAKGC.md        | ComplEx  | oui     | 9.2<br>46.8<br>ComplEx                                                                       |                           |   |
| RAAKGC.md        | ConvE    | oui     | 5.9<br>45.6<br>ConvE                                                                         |                           |   |
| RAAKGC.md        | DistMult | oui     | 9.7<br>44.4<br>DistMult                                                                      |                           |   |
| RAAKGC.md        | HaSa     | oui     | 4.9<br>53.8<br>HaSa                                                                          |                           |   |
| RAAKGC.md        | InsKGC   | oui     | s for the text-based methods, we choose InsKGC (Jiang, Drummond, and Cohn 2023) and Si       |                           |   |
| RAAKGC.md        | KG-BERT  | oui     | ods<br>KG-BERT<br>21.6                                                                       |                           |   |
| RAAKGC.md        | MTL-KGC  | oui     | d<br>MTL-KGC<br>33.1                                                                         |                           |   |
| RAAKGC.md        | QuatE    | oui     | 9.0<br>48.1<br>QuatE                                                                         |                           |   |
| RAAKGC.md        | RAA-KGC  | oui     | d<br>RAA-KGC<br>59.74                                                                        |                           |   |
| RAAKGC.md        | RGCN     | oui     | d<br>RGCN<br>42.7                                                                            |                           |   |
| RAAKGC.md        | RotatE   | oui     | d<br>RotatE<br>47.6                                                                          |                           |   |
| RAAKGC.md        | SimKGC   | oui     | d<br>SimKGC<br>55.31                                                                         |                           |   |
| RAAKGC.md        | StAR     | oui     | d<br>StAR<br>40.1                                                                            |                           |   |
| RAAKGC.md        | TransE   | oui     | ods<br>TransE<br>24.3                                                                        |                           |   |
| RandomCorrupt.md | ComplEx  | oui     | modifications to adapt this task; (2) <b>ComplEx</b> achieves the best performance on all    |                           |   |
| RandomCorrupt.md | DistMult | oui     | ificantly and consistently outperforms <b>DistMult</b> which matches the observations for th |                           |   |

| Article          | Entité    | Trouv é | Extrait (tableau)                                                                         |  |  |
|------------------|-----------|---------|-------------------------------------------------------------------------------------------|--|--|
| RandomCorrupt.md | TransE    | oui     | of link prediction; (2) Surprisingly, <b>TransE</b> has competitive performance with **Co |  |  |
| RCWC.md          | ComplEx   | oui     | 946</td> </tr> <tr> <td>ComplEx [Trouillon et al., 2016]</td> <                           |  |  |
| RCWC.md          | ConvE     | oui     | 947</td> </tr> <tr> <td>ConvE [Dettmers et al., 2018]</td> <t                             |  |  |
| RCWC.md          | DistMult  | oui     | 943</td> </tr> <tr> <td>DistMult [Yang et al., 2014]</td> <td>42                          |  |  |
| RCWC.md          | InteractE | oui     | .54</td> </tr> <tr> <td>InteractE [Vashishth et al., 2020]</td> <                         |  |  |
| RCWC.md          | KGBoost-R | oui     | 951</td> </tr> <tr> <td>KGBoost-R (Ours)</td> <td>16</td>                                 |  |  |
| RCWC.md          | KGBoost-T | oui     | ng></td> </tr> <tr> <td>KGBoost-T (Ours)</td> <td><strong>15</str                         |  |  |
| RCWC.md          | RotatE    | oui     | 956</td> </tr> <tr> <td>RotatE [Sun et al., 2019]</td> <td>40<                            |  |  |
| RCWC.md          | SACN      | oui     | ng></td> </tr> <tr> <td>SACN [Shang et al., 2019]</td> <td>-                              |  |  |
| RCWC.md          | TransE    | oui     | </thead> <tbody> <tr> <td>TransE [Bordes et al., 2013]</td> <td>                          |  |  |
| ReasonKGE.md     | ComplEx   | oui     | </tr> <tr> <td rowspan="3">ComplEx</td> <td>LUBM3U</td> <t                                |  |  |
| ReasonKGE.md     | TransE    | oui     | tbody> <tr> <td rowspan="3">TransE</td> <td>LUBM3U</td> <t                                |  |  |
| RUGA.md          | ComplEx   | oui     | 546</td> </tr> <tr> <td>ComplEx</td> <td>0.690</td> <td>                                  |  |  |
| RUGA.md          | DistMult  | oui     | 468</td> </tr> <tr> <td>DistMult</td> <td>0.634</td> <td>                                 |  |  |
| RUGA.md          | HolE      | oui     | 570</td> </tr> <tr> <td>HolE</td> <td>0.608</td> <td>                                     |  |  |
| RUGA.md          | RUGA      | oui     | </tr> <tr> <td><strong>RUGA</strong></td> <td><strong>0.779                               |  |  |
| RUGA.md          | RUGE      | oui     | 601</td> </tr> <tr> <td>RUGE</td> <td>0.767</td> <td>                                     |  |  |
| RUGA.md          | TransE    | oui     | </thead> <tbody> <tr> <td>TransE</td> <td>h, t</td> <td>                                  |  |  |
| SANS.md          | DistMult  | oui     | </tr> <tr> <td rowspan="6">DistMult</td> <td>KBGAN</td> <td>                              |  |  |
| SANS.md          | RotatE    | oui     | </tr> <tr> <td rowspan="4">RotatE</td> <td>Uniform</td> <                                 |  |  |
| SANS.md          | TransE    | oui     | tbody> <tr> <td rowspan="6">TransE</td> <td>KBGAN (Cai and Wang, 2                        |  |  |
| SelfAdv.md       | ComplEx   | oui     | \overline{\cdot}\$ denotes conjugate for complex vectors, and 2D reshaping for real vect  |  |  |
| SelfAdv.md       | ConvE     | oui     | s, and 2D reshaping for real vectors in ConvE model. TransX represents a wide range o     |  |  |

| Article      | Entité             | Trouvé | Extrait (tableau)                                                                      |                         |         |
|--------------|--------------------|--------|----------------------------------------------------------------------------------------|-------------------------|---------|
| SelfAdv.md   | DistMult           | oui    | $\ln R^k$                                                                              |                         |         |
| SelfAdv.md   | HoIE               | oui    | $\ln C^k$                                                                              |                         |         |
| SelfAdv.md   | pRotatE            | oui    | 0.956                                                                                  |                         |         |
| SelfAdv.md   | RotatE             | oui    | $\ln R^k$                                                                              | $\ln \text{circ r} - t$ | $\ln 2$ |
| SelfAdv.md   | TransE             | oui    | odel. TransX represents a wide range of TransE's variants, such as TransH (Wang et al. |                         |         |
| SNS.md       | DistMult           | oui    | $\ln 2$                                                                                |                         |         |
| SNS.md       | TransH             | oui    |                                                                                        |                         |         |
| SparseNSG.md | ANALOGY            | oui    | 0.1                                                                                    |                         |         |
| SparseNSG.md | ComplEx            | oui    | 0.1                                                                                    |                         |         |
| SparseNSG.md | DistMult           | oui    | 0.01                                                                                   |                         |         |
| SparseNSG.md | TransE             | oui    |                                                                                        |                         |         |
| SparseNSG.md | TransH             | oui    | 0.01                                                                                   |                         |         |
| STC.md       | LFM                | oui    | 1.3                                                                                    |                         |         |
| STC.md       | RESCAL             | oui    |                                                                                        |                         |         |
| STC.md       | SE                 | oui    | 4.1                                                                                    |                         |         |
| STC.md       | SME (bilinear)     | oui    | 0.8                                                                                    |                         |         |
| STC.md       | SME (linear)       | oui    | 9.8                                                                                    |                         |         |
| STC.md       | TKRL (RHE)         | oui    | 7.2                                                                                    |                         |         |
| STC.md       | TKRL (RHE+STC)     | NON    | -                                                                                      |                         |         |
| STC.md       | TKRL (RHE+STC+TCE) | NON    | -                                                                                      |                         |         |
| STC.md       | TKRL (WHE)         | oui    | 9.4                                                                                    |                         |         |
| STC.md       | TKRL (WHE+STC)     | NON    | -                                                                                      |                         |         |
| STC.md       | TKRL (WHE+STC+TCE) | NON    | -                                                                                      |                         |         |
| STC.md       | TransE             | oui    | 3.1                                                                                    |                         |         |

| Article                | Entité                            | Trouvé | Extrait (tableau)                                                             |               |           |
|------------------------|-----------------------------------|--------|-------------------------------------------------------------------------------|---------------|-----------|
| STC.md                 | TransE+STC+TC<br>E                | NON    | -                                                                             |               |           |
| STC.md                 | TransE+TCE                        | NON    | -                                                                             |               |           |
| STC.md                 | TransR                            | oui    | 2.1</td> </tr> <tr> <td>TransR</td><br><td>199</td> <td>7                     |               |           |
| STC.md                 | TransR+STC+TC<br>E                | NON    | -                                                                             |               |           |
| STC.md                 | TransR+TCE                        | NON    | -                                                                             |               |           |
| TANS.md                | ComplEx                           | oui    | <td>FB15k-237</td><br><td>ComplEx</td> <td>25.0</td><br><td>                  |               |           |
| TANS.md                | DistMult                          | oui    | <td>FB15k-237</td><br><td>DistMult</td> <td>24.0</td> <td>                    |               |           |
| TANS.md                | HAKE                              | oui    | <td>FB15k-237</td> <td>HAKE</td><br><td>33.0</td> <td>                        |               |           |
| TANS.md                | HousE                             | oui    | <td>FB15k-237</td> <td>HousE</td><br><td>33.5</td> <td>                       |               |           |
| TANS.md                | RotatE                            | oui    | <td>FB15k-237</td> <td>RotatE</td><br><td>33.0</td> <td>                      |               |           |
| TANS.md                | TransE                            | oui    | <td>FB15k-237</td><br><td>TransE</td> <td>33.0</td> <td>                      |               |           |
| TruncatedNS.md         | IPTransE                          | oui    | 258</td> </tr> <tr><br><td>IPTransE</td> <td>32.73</td><br><td>               |               |           |
| TruncatedNS.md         | JAPE                              | oui    | 412</td> </tr> <tr> <td>JAPE</td><br><td>14.45</td> <td>                      |               |           |
| TruncatedNS.md         | MTransE                           | oui    | 007</td> </tr> <tr> <td>MtransE</td><br><td>16.51</td> <td>                   |               |           |
| TruncatedNS.md         | TransE                            | oui    | </thead> <tbody> <tr><br><td>TransE</td> <td>8.35</td> <td>                   |               |           |
| TruncatedNS.md         | Truncated<br>Negative<br>Sampling | NON    | -                                                                             |               |           |
| TuckerDNCaching.<br>md | ComplEx                           | oui    | ^n\$</td> </tr> <tr> <td>ComplEx</td><br><td>\$Re(h^{\top} \cdot diag(        |               |           |
| TuckerDNCaching.<br>md | DistMult                          | oui    | tic<br/>matching-based</td><br><td>DistMult</td> <td>\$h^{\top} \cdot diag(r) |               |           |
| TuckerDNCaching.<br>md | TransD                            | oui    | ^n\$</td> </tr> <tr> <td>TransD</td><br><td>\$\left\                          | h + w_r w_h^t |           |
| TuckerDNCaching.<br>md | TransE                            | oui    | nal<br/>distance-based</td><br><td>TransE</td> <td>\$\                        | h + r - t\    | _{1/2}\$< |
| TypeAugmented.md       | ComplEx                           | oui    | td> </tr> <tr><br><td>TaKE+ComplEx</td> <td>O(d +<br>k)</td>                  |               |           |
| TypeAugmented.md       | DistMult                          | oui    | td> </tr> <tr><br><td>TaKE+Distmult</td> <td>O(d +<br>k)</td>                 |               |           |
| TypeAugmented.md       | RotatE                            | oui    | td> </tr> <tr> <td>TaKE+RotatE</td><br><td>O(d + k)</td>                      |               |           |
| TypeAugmented.md       | SimpleE                           | oui    | td> </tr> <tr> <td>TaKE+SimpleE</td><br><td>O(d + k)</td>                     |               |           |
| TypeAugmented.md       | TaKE-ComplEx                      | oui    | 428</td> </tr> <tr><br><td>TaKE-ComplEx</td><br><td>0.778</td> <td>           |               |           |

| Article            | Entité                   | Trouvé     | Extrait (tableau)                                                                            |                                                                 |  |
|--------------------|--------------------------|------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--|
| TypeAugmented.md   | TaKE-DistMult            | oui        | 419</td> </tr> <tr><br><td>TaKE-DistMult</td><br><td>0.757</td> <td>                         |                                                                 |  |
| TypeAugmented.md   | TaKE-RotatE              | oui        | 533</td> </tr> <tr><br><td>TaKE-RotatE</td><br><td>0.804</td> <td>                           |                                                                 |  |
| TypeAugmented.md   | TaKE-SimpleE             | oui        | 476</td> </tr> <tr><br><td>TaKE-SimpleE</td><br><td><strong>0.806</strong>                   |                                                                 |  |
| TypeAugmented.md   | TaKE-TransE              | oui        | 465</td> </tr> <tr><br><td>TaKE-TransE</td><br><td>0.490</td> <td>                           |                                                                 |  |
| TypeAugmented.md   | TransE                   | oui        | td> </tr> <tr> <td>TaKE+TransE</td><br><td>O(d + k)</td>                                     |                                                                 |  |
| TypeAugmented.md   | TypeComplex              | oui        | R\                                                                                           | )</td> </tr> <tr> <td><br>TypeComplex</td><br><td>O(d + k)</td> |  |
| TypeAugmented.md   | TypeDM                   | oui        | </thead> <tbody> <tr><br><td>TypeDM</td> <td>O(d + k)</td>                                   |                                                                 |  |
| TypeConstraints.md | mwNN                     | oui        | omparison of AUPRC and AUROC<br>result for mwNN [8] with and without<br>exploiting prior k   |                                                                 |  |
| TypeConstraints.md | RESCAL                   | oui        | omparison of AUPRC and AUROC<br>result for RESCAL with and without<br>exploiting prior knowl |                                                                 |  |
| TypeConstraints.md | TransE                   | oui        | omparison of AUPRC and AUROC<br>result for TransE with and without<br>exploiting prior knowl |                                                                 |  |
| Uniform.md         | LFM                      | oui        | .06</td> </tr> <tr> <td>LFM [6]</td><br><td>\$O(n_e k + n_r k +                              |                                                                 |  |
| Uniform.md         | RESCAL                   | oui        | .75</td> </tr> <tr> <td>RESCAL<br>[11]</td> <td>\$O(n_e k + n_r k^                           |                                                                 |  |
| Uniform.md         | SME(bilinear)            | oui        | .82</td> </tr> <tr><br><td>SME(BILINEAR) [2]</td><br><td>\$O(n_e k + n_r k                   |                                                                 |  |
| Uniform.md         | SME(linear)              | oui        | .47</td> </tr> <tr> <td>SME(LINEAR)<br>[2]</td> <td>\$O(n_e k + n_r k                        |                                                                 |  |
| Uniform.md         | Structured<br>Embeddings | <b>NON</b> | -                                                                                            |                                                                 |  |
| Uniform.md         | TransE                   | oui        | .84</td> </tr> <tr> <td>TransE</td><br><td>\$O(n_e k + n_r k)\$</td>                         |                                                                 |  |
| Uniform.md         | Unstructured             | oui        | </thead> <tbody> <tr><br><td>Unstructured [2]</td><br><td>\$O(n_e k)\$</td>                  |                                                                 |  |

## B. Preuve - MENTIONNÉS seulement (235)

| Article         | Entité | Trouvé | Extrait (prose)                                                                              |
|-----------------|--------|--------|----------------------------------------------------------------------------------------------|
| AdaptativeNS.md | HAKE   | oui    | relationships in knowledge<br>graphs. The HAKE [20] model<br>is designed for graph embedd    |
| AdaptativeNS.md | MEI    | oui    | ent intricacies of the<br>knowledge graph. MEI [13]<br>and MEIM [14] partition each<br>embed |

| Article         | Entité   | Trouvé | Extrait (prose)                                                                           |
|-----------------|----------|--------|-------------------------------------------------------------------------------------------|
| AdaptativeNS.md | MEIM     | oui    | es of the knowledge graph. MEI [13] and MEIM [14] partition each embedding into mult      |
| AdaptativeNS.md | RESCAL   | oui    | ressed in vector space representations. RESCAL [10] introduced a tensor factorization     |
| AdaptativeNS.md | TransD   | oui    | between entities and relationships. The TransD [18] model further improves upon TransR    |
| AdaptativeNS.md | TransG   | oui    | and relationships more effectively. The TransG [17] model proposes a solution to the c    |
| AdaptativeNS.md | TransH   | oui    | many and many-to-one relationships, the TransH [16] model extends it by introducing re    |
| AdaptativeNS.md | TransR   | oui    | omes the shortcomings of TransE. In the TransR [7] model, a relation mapping matrix is    |
| ADNS.md         | Complex  | oui    | t years. e.g, TransE [2], DistMult [3], Complex[4], TransH [5], RotatE [6] etc. Almost    |
| ADNS.md         | RotatE   | oui    | , DistMult [3], Complex[4], TransH [5], RotatE [6] etc. Almost all of these models dep    |
| ADNS.md         | TransH   | oui    | , TransE [2], DistMult [3], Complex[4], TransH [5], RotatE [6] etc. Almost all of thes    |
| BatchNS.md      | DistMult | oui    | n operators allows PBG to train RESCAL, DistMult, TransE, and Complex models (Nickel et   |
| BatchNS.md      | word2vec | oui    | this literature are algorithms such as word2vec which allowed word embedding methods to   |
| CAKE.md         | Complex  | oui    | se together with the characteristics of complex relations. Furthermore, a multi-view li   |
| CAKE.md         | DistMult | oui    | osition-based** score function, such as DistMult (Yang et al., 2015): $E(h, r, t) = \sum$ |
| CAKE.md         | JOIE     | oui    | hand, although some KGE models such as JOIE (Hao et al., 2019) employ the ontology        |
| CAKE.md         | RESCAL   | oui    | s such as TransE (Bordes et al., 2013), RESCAL (Nickel et al., 2011), Complex (Trouill    |
| CAKE.md         | TransH   | oui    | amely 1-1, 1-N, N-1, and N-N defined in TransH (Wang et al., 2014) for negative sampli    |



| Article | Entité    | Trouvé | Extrait (prose)                                                                          |
|---------|-----------|--------|------------------------------------------------------------------------------------------|
| CANS.md | GTrans    | oui    | iple as a manifold rather than a point. GTrans [20] used dynamic and static weighting    |
| CANS.md | ManifoldE | oui    | ctors of entities into various spaces. ManifoldE [19] embedded a triple as a manifold ra |
| CANS.md | puTransE  | oui    | d the noise from other relation spaces. puTransE [21] generated multiple embedding space |
| CANS.md | TransD    | oui    | ame vector space. Then TransR [17], and TransD [18], extended TransE, by projecting th   |
| CANS.md | TransH    | oui    | th 1-to-N, N-to-1, or N-to-N relations. TransH [10] proposed to project the entity to    |
| CANS.md | TransR    | oui    | ons were in the same vector space. Then TransR [17], and TransD [18], extended TransE,   |
| CCS.md  | HAKE      | oui    | on translation of the vectors in space. HAKE [31] maps the entities into spatial pol     |
| CCS.md  | NTN       | oui    | SME) [16] model, neural tensor network (NTN) [17] model, and matrix factorization (      |
| CCS.md  | OTE       | oui    | ntities into spatial polar coordinates. OTE [32] exploits the orthogonality relatio      |
| CCS.md  | PTransE   | oui    | TransE [26], TransA [27], TransAt [28], PTransE [29], etc. Recently, RotatE [30] has ch  |
| CCS.md  | RESCAL    | oui    | ) [17] model, and matrix factorization (RESCAL) [18] model. However, these KGE models    |
| CCS.md  | RotatE    | oui    | nsAt [28], PTransE [29], etc. Recently, RotatE [30] has changed the translation proces   |
| CCS.md  | SE        | oui    | tion, such as the structured embedding (SE) [15] model, semantic matching energy (       |
| CCS.md  | SME       | oui    | ) [15] model, semantic matching energy (SME) [16] model, neural tensor network (NTN      |
| CCS.md  | STransE   | oui    | nsD [23], TransM [24], TranSparse [25], STransE [26], TransA [27], TransAt [28], PTrans  |
| CCS.md  | TransA    | oui    | sM [24], TranSparse [25], STransE [26], TransA [27], TransAt [28], PTransE [29], etc.    |

| Article            | Entité     | Trouvé | Extrait (prose)                                                                                |
|--------------------|------------|--------|------------------------------------------------------------------------------------------------|
| CCS.md             | TransAt    | oui    | Sparse [25], STransE [26], TransA [27], TransAt [28], PTransE [29], etc. Recently, Rota        |
| CCS.md             | TransM     | oui    | TransH [21], TransR [22], TransD [23], TransM [24], TranSparse [25], STransE [26], Tr          |
| CCS.md             | TranSparse | oui    | TransR [22], TransD [23], TransM [24], TranSparse [25], STransE [26], TransA [27], TransA      |
| CondConstraints.md | RESCAL     | oui    | nly explores their applicability to the RESCAL model [2]. On the other hand, the work          |
| CondConstraints.md | TransH     | oui    | cheme was introduced by Wang et al. for TransH and is sometimes called the Bernoulli t         |
| CondConstraints.md | TRESCAL    | oui    | exploiting type information [6]. While TRESCAL explicitly tries to make use of type co         |
| dans.md            | RGCN       | oui    | relational graph convolutional network (RGCN) [26] to serve as our base embedding mo           |
| dans.md            | TransE     | oui    | xt{China} \rangle\$ and a classic method TransE [2]. TransE maps each entity and relati        |
| DeMix.md           | TransH     | oui    | amely 1-N, N-1, 1-1, and N-N defined in TransH [24], we record the num of positive tri         |
| DHNS.md            | KGDM       | oui    | idering the query node's context, while KGDM [16] applies DDPM to estimate the proba           |
| DHNS.md            | LAFA       | oui    | y in an adversarial training framework. LAFA [24] considers the relationships between          |
| DHNS.md            | QuatE      | oui    | ntisymmetric relations. RotatE [25] and QuatE [36] introduce rotational and quaternion         |
| DHNS.md            | RESCAL     | oui    | ar interaction-based KGE models such as RESCAL [22] and DistMult [35], formulated as:          |
| DHNS.md            | TransH     | oui    | based KGE models such as TransE [3] and TransH [31], formulated as: $\mathbf{C}(\mathbf{x}_e,$ |
| DHNS.md            | VISTA      | oui    | aggregation of multi-modal information. VISTA [11] designs three transformer-based en          |
| DNS.md             | ComplEx    | oui    | ued embeddings into their corresponding complex variants, and performs Hermitian dot pr        |
| DNS.md             | HoIE       | oui    | teractions within the triples for their HoIE system. Trouillon et al. [2016] represe           |

| Article          | Entité                | Trouvé | Extrait (prose)                                                                                                 |
|------------------|-----------------------|--------|-----------------------------------------------------------------------------------------------------------------|
| DNS.md           | Neural Tensor Network | oui    | h as <i>RESCAL</i> Nickel et al. [2011] and <i>Neural Tensor Network</i> Socher et al. [2013] employ higher ord |
| DNS.md           | TransH                | oui    | Wang et al. [2014] define an algorithm <i>TransH</i> that behaves like <i>TransE</i> with a mino                |
| EANS.md          | RESCAL                | oui    | relation vectors into various spaces. RESCAL (Nickel et al., 2011), DistMult (Yang e                            |
| EANS.md          | TransH                | oui    | . Various extensions of TransE, such as TransH (Wang et al., 2014), TransR (Lin et al.                          |
| EANS.md          | TransR                | oui    | sE, such as TransH (Wang et al., 2014), TransR (Lin et al., 2015) and TransD (Ji et al                          |
| ERDNS.md         | CompGCN               | oui    | ], ConvE [5], and InteractE [24], while CompGCN [5] is a generalized version of several                         |
| ERDNS.md         | ConvKB                | oui    | odels. Examples of CNN-based models are ConvKB [18], ConvE [5], and InteractE [24], wh                          |
| ERDNS.md         | InteractE             | oui    | models are ConvKB [18], ConvE [5], and InteractE [24], while CompGCN [5] is a generalize                        |
| ERDNS.md         | SimpleE               | oui    | ples and lower scores to negative ones. Simple negative sampling methods like uniform                           |
| ERDNS.md         | TransD                | oui    | amples of TD models include TransE [2], TransD [8], TransR [15], and RotatE [21], whil                          |
| ERDNS.md         | TransR                | oui    | models include TransE [2], TransD [8], TransR [15], and RotatE [21], while popular SM                           |
| eTruncatedUNS.md | KR-EAR                | oui    | tes reverse triples and relation paths. KR-EAR [Lin et al., 2016] introduces categoric                          |
| eTruncatedUNS.md | PTransE               | oui    | 2014], TransR [Lin et al., 2015b] and PTransE [Lin et al., 2015a]. Most existing KG                             |
| eTruncatedUNS.md | TransE                | oui    | nherent KG semantics. As an early work, TransE [Bordes et al., 2013] interprets a rela                          |
| eTruncatedUNS.md | TransH                | oui    | further improved by many works, such as TransH [Wang et al., 2014], TransR [Lin et al                           |
| eTruncatedUNS.md | TransR                | oui    | s, such as TransH [Wang et al., 2014], TransR [Lin et al., 2015b] and PTransE [Lin et                           |
| GHN.md           | ComplEx               | oui    | al., 2013), TransH (Wang et al., 2014), Complex (Trouillon et al., 2016), and RotatE (S                         |
| GHN.md           | TransH                | oui    | s include TransE (Bordes et al., 2013), TransH (Wang et al., 2014), Complex (Trouillon                          |

| Article     | Entité   | Trouvé | Extrait (prose)                                                                          |
|-------------|----------|--------|------------------------------------------------------------------------------------------|
| GibbsNS.md  | KG-BERT  | oui    | ed pre-trained models to tackle issues. KG-BERT [14], for instance, is a BERT-based kno  |
| GibbsNS.md  | RESCAL   | oui    | . An eminent contender in this class is RESCAL proposed by Nickel et al. [24], which c   |
| GibbsNS.md  | TransH   | oui    | gh similarity. To overcome challenges, TransH, introduced by [9], utilizes hyperplane    |
| GNDN.md     | DistMult | oui    | ions over entities and relations, and a DistMult based decoder for factorization. The GC |
| GNDN.md     | ELECTRA  | oui    | ecently developed language model called ELECTRA [37], which is a compute-efficient GAN-  |
| GNDN.md     | HypER    | oui    | igm and reports better overall results. HypER [26] uses a 1D-relation-specific convol    |
| GNDN.md     | KCGAN    | oui    | iscriminative belief prediction models. KCGAN thus invokes a game between generator-n    |
| GNDN.md     | KG-BERT  | oui    | ansformer-based encoding models such as KG-BERT [35] are inspired by transformer-based   |
| GNDN.md     | PTransE  | oui    | red as PathG-RNNs. We also compare with PTransE which is a variant of TransE for integr  |
| GNS.md      | BERT     | oui    | state-of-the-art embedding method - KG-BERT for triple classification task - on a b      |
| GNS.md      | ComplEx  | oui    | n KGs such as: TransE [5], Rescal [12], ComplEx [16], DistMult [20] and KG-BERT [21], a  |
| GNS.md      | DistMult | oui    | TransE [5], Rescal [12], ComplEx [16], DistMult [20] and KG-BERT [21], and all these me  |
| GNS.md      | Rescal   | oui    | embeddings in KGs such as: TransE [5], Rescal [12], ComplEx [16], DistMult [20] and K    |
| GNS.md      | TransE   | oui    | and relation embeddings in KGs such as: TransE [5], Rescal [12], ComplEx [16], DistMul   |
| GNS.md      | TransOWL | oui    | e sampling for triple classification 5 TransOWL [7] leverages ontology axioms in order   |
| GraphGAN.md | PPNE     | oui    | nder the supervision of edge existence. PPNE (Li et al. 2017b) directly learns verte     |
| GraphGAN.md | SDNE     | oui    | aining data in the graph. For instance, SDNE (Wang, \*M. Guo is the corresponding a      |

| Article  | Entité                | Trouvé | Extrait (prose)                                                                                                                   |
|----------|-----------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------|
| IGAN.md  | Neural Tensor Network | oui    | $g(g)$ where $g()$ is tanh function. <b>Neural Tensor Network</b> (Socher et al. 2013) extends Single L                           |
| KBGAN.md | ConvE                 | oui    | o combine the two entities in a triple. CONVE (Dettmers et al., 2017) uses a convolut                                             |
| KBGAN.md | HolE                  | oui    | iple as a manifold rather than a point. HOLE (Nickel et al., 2016) employs circular                                               |
| KBGAN.md | ManifoldE             | oui    | ent models achieve strong performances. MANIFOLDE (Xiao et al., 2016) embeds a triple as                                          |
| KBGAN.md | RESCAL                | oui    | graph embedding (KGE) techniques (e.g., RESCAL (Nickel et al., 2011), TRANSE (Bordes e                                            |
| KBGAN.md | TransH                | oui    | ased embedding. Later variants, such as TRANSH (Wang et al., 2014), TRANSR (Lin et al.                                            |
| KBGAN.md | TransR                | oui    | ts, such as TRANSH (Wang et al., 2014), TRANSR (Lin et al., 2015) and TRANSD (Ji et al                                            |
| KSGAN.md | DistMult              | oui    | ng model (e.g., TransE [6], TransD [7], DistMult [8], ComplEx [9]). However, previous w                                           |
| KSGAN.md | GTrans                | oui    | (or $\mathbf{t} - \mathbf{r}$ ). As a generic model, GTrans [17] introduces eigenstate and mimesis                                |
| KSGAN.md | ManifoldE             | oui    | ict magnitude constraints between them. ManifoldE [16] uses a manifold function to constr                                         |
| KSGAN.md | MLP                   | oui    | ic matching models such as NTN [21] and MLP [22], focus on neural network architect                                               |
| KSGAN.md | NTN                   | oui    | Other semantic matching models such as NTN [21] and MLP [22], focus on neural netw                                                |
| KSGAN.md | RESCAL                | oui    | from the translational distance model, RESCAL [18] is a classic semantic matching mod                                             |
| KSGAN.md | SimpleE               | oui    | complex space rather than a real space. SimpleE [19] simplifies ComplEx by considering                                            |
| KSGAN.md | TransF                | oui    | ons. Targeting at more flexible models, TransF [15] ensures that $\mathbf{t}$ (or $\mathbf{h}$ ) has                              |
| KSGAN.md | TransH                | oui    | $\mathbf{t} \cdot (\mathbf{h}, \mathbf{r}, \mathbf{t})$ . Various variants such as TransH [12], TransM [13], TransR [14] and Tran |
| KSGAN.md | TransM                | oui    | . Various variants such as TransH [12], TransM [13], TransR [14] and TransD [7] have b                                            |

| Article  | Entité   | Trouvé | Extrait (prose)                                                                           |
|----------|----------|--------|-------------------------------------------------------------------------------------------|
| KSGAN.md | TransR   | oui    | models such as TransH [12], TransM [13], TransR [14] and TransD [7] have been developed   |
| KSGAN.md | TuckER   | oui    | different similarity scoring function. TuckER [20] is based on Tucker decomposition       |
| LAS.md   | ANALOGY  | oui    | ability-based models, e.g. ProjE [21], ANALOGY [22] and R-GCN [28], also achieve good     |
| LAS.md   | DKRL     | oui    | ] exploit relation paths, TEKE [33] and DKRL [34] embed knowledge graph with textual      |
| LAS.md   | HoIE     | oui    | models (e.g., RESCAL [17], DistMult [18], HoIE [19], ComplEx [20], and ConvE [23] etc)    |
| LAS.md   | ITransF  | oui    | [13], STransE [14], puTransE [15], and ITransF [16]) define an energy function according  |
| LAS.md   | KALE     | oui    | knowledge graph with textual information, KALE [35] and RUGE [36] utilize logical rule    |
| LAS.md   | KG2E     | oui    | 6], TransH [7], TransR [8], TransD [9], KG2E [10], TransA [11], TransSparse [12], Tra     |
| LAS.md   | ProjE    | oui    | ny other probability-based models, e.g. ProjE [21], ANALOGY [22] and R-GCN [28], also     |
| LAS.md   | PTransE  | oui    | and TKRL [30] consider the entity type, PTransE [31] and RTransE [32] exploit relation    |
| LAS.md   | puTransE | oui    | Sparse [12], TransG [13], STransE [14], puTransE [15], and ITransF [16]) define an energy |
| LAS.md   | RESCAL   | oui    | models Probability-based models (e.g., RESCAL [17], DistMult [18], HoIE [19], ComplEx     |
| LAS.md   | RTransE  | oui    | sider the entity type, PTransE [31] and RTransE [32] exploit relation paths, TEKE [33]    |
| LAS.md   | RUGE     | oui    | with textual information, KALE [35] and RUGE [36] utilize logical rules. In this pa       |
| LAS.md   | SSE      | oui    | further improve the task. For example, SSE [29] and TKRL [30] consider the entity         |
| LAS.md   | STransE  | oui    | nsA [11], TransSparse [12], TransG [13], STransE [14], puTransE [15], and ITransF [16])   |
| LAS.md   | TEKE     | oui    | nd RTransE [32] exploit relation paths, TEKE [33] and DKRL [34] embed knowledge grap      |

| Article           | Entité                                  | Trouvé     | Extrait (prose)                                                                                                       |
|-------------------|-----------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------|
| LAS.md            | TKRL                                    | oui        | ove the task. For example, SSE [29] and TKRL [30] consider the entity type, PTransE                                   |
| LAS.md            | TransA                                  | oui        | [7], TransR [8], TransD [9], KG2E [10], TransA [11], TranSparse [12], TransG [13], STr                                |
| LAS.md            | TransG                                  | oui        | G2E [10], TransA [11], TranSparse [12], TransG [13], STransE [14], puTransE [15], and                                 |
| LAS.md            | TransH                                  | oui        | slation-based models (e.g., TransE [6], TransH [7], TransR [8], TransD [9], KG2E [10],                                |
| LAS.md            | TranSparse                              | oui        | 8], TransD [9], KG2E [10], TransA [11], TranSparse [12], TransG [13], STransE [14], puTran                            |
| LAS.md            | TransR                                  | oui        | d models (e.g., TransE [6], TransH [7], TransR [8], TransD [9], KG2E [10], TransA [11]                                |
| Localcognitive.md | Relation-adaptive translating Embedding | oui        | 10.04863v1 [cs.CL] 10 Oct 2020 # RatE: Relation-Adaptive Translating Embedding for Knowledge Graph Completion **Hao H |
| Localcognitive.md | TorusE                                  | <b>NON</b> | -                                                                                                                     |
| Localcognitive.md | TransD                                  | <b>NON</b> | -                                                                                                                     |
| Localcognitive.md | TransH                                  | <b>NON</b> | -                                                                                                                     |
| Localcognitive.md | TransR                                  | <b>NON</b> | -                                                                                                                     |
| LTS.md            | GAN                                     | oui        | c distribution sampling, represented by GAN[2], is used to provide high quality neg                                   |
| MCNS.md           | FastGCN                                 | oui        | lleviate the receptive field expansion. FastGCN [5] adopts importance sampling in each                                |
| MCNS.md           | LINE                                    | oui        | Embedding methods (e.g. DeepWalk [27], LINE [32]) and Graph Neural Networks (e.g. G                                   |
| MCNS.md           | node2vec                                | oui        | s, such as DeepWalk [27], LINE [32] and node2vec [11], actually assign each node two emb                              |
| NMiss.md          | ComplEx                                 | oui        | methods - RESCAL, TransE, DistMult and ComplEx - and evaluate on benchmark datasets -                                 |
| NMiss.md          | HolE                                    | oui        | s as well as the Holographic Embedding (HolE) model, so HolE was not included<sup>1<                                  |
| NMiss.md          | Neural Tensor Networks                  | oui        | ns. Methods such as <b>Rescal</b> [21] and Neural Tensor Networks [24] learn millions of parameters that              |

| Article      | Entité     | Trouvé | Extrait (prose)                                                                           |
|--------------|------------|--------|-------------------------------------------------------------------------------------------|
| NoiGAN.md    | CKRL       | oui    | 19)), (3) noise aware KGE models (e.g., CKRL (Xie et al., 2018)) and (4) KGE models       |
| NoiGAN.md    | ComplEx    | oui    | 2011), DistMult (Yang et al., 2014) and ComplEx (Trouillon et al., 2016). To optimize t   |
| NoiGAN.md    | RESCAL     | oui    | esentations. The typical models include RESCAL (Nickel et al., 2011), DistMult (Yang e    |
| NoiGAN.md    | TransH     | oui    | relation. TransE (Bordes et al., 2013), TransH (Wang et al., 2014) and TransR (Lin et     |
| NoiGAN.md    | TransR     | oui    | , 2013), TransH (Wang et al., 2014) and TransR (Lin et al., 2015) are the representati    |
| NSCaching.md | ANALOGY    | oui    | DistMult [46], HoIE [31], ComplEx [38], ANALOGY [27], etc. All these methods are summar   |
| NSCaching.md | HoIE       | oui    | e variants of RESCAL are DistMult [46], HoIE [31] and ComplEx [38]. DistMult simplif      |
| NSCaching.md | ManifoldE  | oui    | nsD [20], TranSparse [21], TransM [11], ManifoldE [45], etc., and semantic matching model |
| NSCaching.md | RESCAL     | oui    | e plausibility of triplets $(h, r, t)$ . RESCAL [32] is the most original model. The en   |
| NSCaching.md | TransM     | oui    | nsR [26], TransD [20], TranSparse [21], TransM [11], ManifoldE [45], etc., and semanti    |
| NSCaching.md | TranSparse | oui    | TransH [42], TransR [26], TransD [20], TranSparse [21], TransM [11], ManifoldE [45], etc. |
| NSCaching.md | TransR     | oui    | his problem, variants like TransH [42], TransR [26], TransD [20] are introduced to pro    |
| PNS.md       | TransG     | oui    | o the current state-of-the-art approach TransG[4] to further improve its performance.     |
| RAAKGC.md    | BERTRL     | oui    | lation-aware neighborhood of the query. BERTRL (Zha, Chen, and Yan 2022) collects poss    |
| RAAKGC.md    | ConKGC     | oui    | he query as extra context to the model. ConKGC (Shang et al. 2023) introduces a sampli    |
| RAAKGC.md    | DKRL       | oui    | . 2024). The pioneer text-based method, DKRL (Xie et al. 2016), initially integrates      |
| RAAKGC.md    | GenKGC     | oui    | sion. Besides the mentioned algorithms, GenKGC (Xie et al. 2022) and KICGPT (Wei et al    |



| Article          | Entité        | Trouvé | Extrait (prose)                                                                                           |
|------------------|---------------|--------|-----------------------------------------------------------------------------------------------------------|
| RAAKGC.md        | KEPLER        | oui    | ng KG-BERT (Yao, Mao, and Luo 2019) and KEPLER (Wang et al. 2021b), utilize a Pre-trai                    |
| RAAKGC.md        | KG-S2S        | oui    | KG-T5-context (Kochsiek et al. 2023) and KG-S2S (Chen et al. 2022a) model the link pred                   |
| RAAKGC.md        | KG-T5-context | oui    | r encoding text descriptions. Moreover, KG-T5-context (Kochsiek et al. 2023) and KG-S2S (Chen             |
| RAAKGC.md        | KICGPT        | oui    | gorithms, GenKGC (Xie et al. 2022) and KICGPT (Wei et al. 2023) go beyond the neighbo                     |
| RAAKGC.md        | LMKE          | oui    | s by the decoder directly. Furthermore, LMKE (Wang et al. 2022b) introduces a contra                      |
| RandomCorrupt.md | ConvE         | oui    | combine the two entities in a triple. <b>ConvE</b> (Dettmers et al., 2017) uses a convol                  |
| RandomCorrupt.md | HOLE          | oui    | <b>DistMult</b> into the complex space. <b>HOLE</b> (Nickel et al., 2016) employs circula                 |
| RandomCorrupt.md | ManifoldE     | oui    | tors of entities into various spaces. <b>ManifoldE</b> (Xiao et al., 2016) embeds a triple a              |
| RandomCorrupt.md | PTransE       | oui    | ), <b>TransR</b> (Lin et al., 2015b) and <b>PTransE</b> (Lin et al., 2015a). However, limited             |
| RandomCorrupt.md | RESCAL        | oui    | le as a manifold rather than a point. <b>RESCAL</b> (Nickel et al., 2011) is one of the e                 |
| RandomCorrupt.md | TransD        | oui    | ), <b>TransR</b> (Lin et al., 2015b) and <b>TransD</b> (Ji et al., 2015), extend <b>TransE</b>            |
| RandomCorrupt.md | TransH        | oui    | ed embedding. Later variants, such as <b>TransH</b> (Wang et al., 2014), <b>TransR</b> (Lin               |
| RandomCorrupt.md | TransR        | oui    | ch as <b>TransH</b> (Wang et al., 2014), <b>TransR</b> (Lin et al., 2015b) and <b>TransD</b> (J           |
| RCWC.md          | RESCAL        | oui    | trix $\$M_r\$$ is used to model a relation. RESCAL [Nickel et al., 2011] suffers from mode                |
| RCWC.md          | TransH        | oui    | $\$r \approx 0\$$ for symmetric relations. TransH [Lin et al., 2015] and TransR [Lin et a                 |
| RCWC.md          | TransR        | oui    | elations. TransH [Lin et al., 2015] and TransR [Lin et al., 2015] model symmetric rela                    |
| ReasonKGE.md     | Embed2Reason  | oui    | mpling. For example, a related method <i>Embed2Reason</i> ( <i>E2R</i> ) has been proposed by Garg *et al |
| RUGA.md          | RESCAL        | oui    | coring function. Typical models include RESCAL <sup>[13]</sup> , DistMult <sup>[14]</sup>                 |

| Article      | Entité     | Trouvé | Extrait (prose)                                                                                  |
|--------------|------------|--------|--------------------------------------------------------------------------------------------------|
| RUGA.md      | TransD     | oui    | H <sup>[9]</sup> , TransR <sup>[10]</sup> , TransD <sup>[11]</sup> and so on. As shown in F      |
| RUGA.md      | TransH     | oui    | ne to expand and apply TranE, including TransH <sup>[9]</sup> , TransR <sup>[10]</sup> , T       |
| RUGA.md      | TransR     | oui    | TranE, including TransH <sup>[9]</sup> , TransR <sup>[10]</sup> , TransD <sup>[11]</sup> a       |
| SelfAdv.md   | KG2E       | oui    | We also compare an additional approach KG2E_KL (He et al., 2015), which is a probab              |
| SelfAdv.md   | STransE    | oui    | 2014), TransR (Lin et al., 2015b), and STransE (Nguyen et al., 2016), where $g_{r,i}(\backslash$ |
| SelfAdv.md   | TorusE     | oui    | and concurrent work to our work is the TorusE (Ebisu & Ichise, 2018) model, which def            |
| SelfAdv.md   | TransH     | oui    | ide range of TransE's variants, such as TransH (Wang et al., 2014), TransR (Lin et al.           |
| SelfAdv.md   | TransR     | oui    | ts, such as TransH (Wang et al., 2014), TransR (Lin et al., 2015b), and STransE (Nguye           |
| SparseNSG.md | NTN        | oui    | other kinds of models, such as SE, SME, NTN. However, a practical knowledge graph i              |
| SparseNSG.md | SE         | oui    | 15], and other kinds of models, such as SE, SME, NTN. However, a practical knowled               |
| SparseNSG.md | SME        | oui    | ter contains, Semantic Matching Energy (SME) [10], Semantically Smooth Embedding (S              |
| SparseNSG.md | SSE        | oui    | E) [10], Semantically Smooth Embedding (SSE) [11], and translation-based methods [1              |
| SparseNSG.md | TransA     | oui    | sing, some presentation models, such as TransA [16], TransG [19], replace the relation           |
| SparseNSG.md | TransD     | oui    | ls, such as TransE, TransH, TransR, and TransD [15], and other kinds of models, such a           |
| SparseNSG.md | TransG     | oui    | esentation models, such as TransA [16], TransG [19], replace the relation of a triplet           |
| SparseNSG.md | TranSparse | oui    | ment, widely used in TransR, TransD and TranSparse [18], can effectively reduce the probab       |
| SparseNSG.md | TransR     | oui    | s, such as TransE [12], TransH [13] and TransR [14], are the most used knowledge graph           |

| Article        | Entité                                          | Trouvé | Extrait (prose)                                                                                                            |
|----------------|-------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------|
| STC.md         | DKRL                                            | oui    | s, entity names or entity descriptions. DKRL [Xie <i>et al.</i> , 2016] proposes descripti                                 |
| STC.md         | NTN                                             | oui    | triples, is significant for RL in KGs. NTN [Socher <i>et al.</i> , 2013] encodes 3-way t                                   |
| STC.md         | SME                                             | oui    | 2011; 2012], SE [Bordes <i>et al.</i> , 2011], SME [Bordes <i>et al.</i> , 2012; 2014] and LFM [Je                         |
| STC.md         | TransD                                          | oui    | judging the distance between entities. TransD [Ji <i>et al.</i> , 2015] proposes dynamic ma                                |
| STC.md         | TransH                                          | oui    | o-1 and N-to-N. To address this issue, TransH [Wang <i>et al.</i> , 2014b] interprets relat                                |
| STC.md         | Type-embodied Knowledge Representation Learning | oui    | paper, we propose a novel method named Type-embodied Knowledge Representation Learning (TKRL) to take advantages of hie... |
| TANS.md        | BERT                                            | oui    | s, Clark <i>et al.</i> (2020a) indicate that a BERT (Devlin <i>et al.</i> , 2019)-like model ELECTR                        |
| TANS.md        | ELECTRA                                         | oui    | a BERT (Devlin <i>et al.</i> , 2019)-like model ELECTRA (Clark <i>et al.</i> , 2020b) uses the NS loss                     |
| TANS.md        | GenKGC                                          | oui    | ods like KGT5 (Saxena <i>et al.</i> , 2022) and GenKGC (Xie <i>et al.</i> , 2022) are unique in direct                     |
| TANS.md        | KEPLER                                          | oui    | guage model (PLM)-based approaches like KEPLER (Wang <i>et al.</i> , 2021) and SimKGC (Wang <i>et</i>                      |
| TANS.md        | KGT5                                            | oui    | ning. Generation-based KGC methods like KGT5 (Saxena <i>et al.</i> , 2022) and GenKGC (Xie <i>e</i>                        |
| TANS.md        | SimKGC                                          | oui    | hes like KEPLER (Wang <i>et al.</i> , 2021) and SimKGC (Wang <i>et al.</i> , 2022) also have an import                     |
| TANS.md        | word2vec                                        | oui    | n <i>et al.</i> (2019) import subsampling from word2vec (Mikolov <i>et al.</i> , 2013) to KGE. Subsampl                    |
| TruncatedNS.md | BootEA                                          | oui    | ame, relationship, and attribute views. BootEA [24] proposes iterative entity alignmen                                     |
| TruncatedNS.md | CrossE                                          | oui    | sed in the entity triple, respectively. CrossE [13] believes that bidirectional effect                                     |
| TruncatedNS.md | DAT                                             | oui    | y updated using newly aligned entities. DAT [21] encodes both entity names and enti                                        |
| TruncatedNS.md | HAKE                                            | oui    | modelled entities in hyperbolic space. HAKE [16] believes that polar coordinates ar                                        |

| Article            | Entité    | Trouvé     | Extrait (prose)                                                                          |
|--------------------|-----------|------------|------------------------------------------------------------------------------------------|
| TruncatedNS.md     | KD-CoE    | oui        | been abstracted to related data types. KD-CoE [26] jointly learns multi-lingual entit    |
| TruncatedNS.md     | MultiKE   | oui        | function of the interaction embeddings. MultiKE [23] uses multiple views of entities to  |
| TruncatedNS.md     | MuRP      | oui        | d as rotation changes in complex space. MuRP [15] points out that the relationships      |
| TruncatedNS.md     | RESCAL    | oui        | ions to gauge the rationality of facts. RESCAL [11] was the earliest semantic matching   |
| TruncatedNS.md     | RotatE    | oui        | ntities and relationships. According to RotatE [14], which can model and infer differe   |
| TruncatedNS.md     | SimpleE   | oui        | ons between latent factors of entities. SimpleE [12] restricts the relationship matrix   |
| TruncatedNS.md     | TransD    | oui        | wo different projection matrices by the TransD [17] model, respectively. According to    |
| TruncatedNS.md     | TransEdge | oui        | erforms well in handling tail entities. TransEdge [22] interacts with embeddings between |
| TruncatedNS.md     | TransH    | oui        | n the corresponding relationship space. TransH [19] thinks an entity can have many rep   |
| TruncatedNS.md     | TransR    | oui        | [17] model, respectively. According to TransR [18], each entity is made up of several    |
| TuckerDNCaching.md | RESCAL    | <b>NON</b> | -                                                                                        |
| TuckerDNCaching.md | RotatE    | oui        | ich has an impact on efficiency. Later, RotatE (Sun et al., 2019) introduced a self-ad   |
| TuckerDNCaching.md | TorusE    | oui        | lational distance-based models, such as TorusE (Ebisu et al., 2018), by addressing the   |
| TuckerDNCaching.md | TuckER    | oui        | ng: high-quality negative sampling with tucker decomposition Tiroshan Madushanka<sup>    |
| TypeAugmented.md   | HAKE      | oui        | g to tail embedding. Soon after RotatE, HAKE<sup>16</sup> and RatE<sup>17</sup> make     |
| TypeAugmented.md   | JOIE      | oui        | ons to model the semantic transitivity. JOIE<sup>24</sup> directly represents a KG a     |
| TypeAugmented.md   | RatE      | oui        | oon after RotatE, HAKE<sup>16</sup> and RatE<sup>17</sup> make slight improvements t     |

| Article            | Entité                | Trouvé | Extrait (prose)                                                                                       |
|--------------------|-----------------------|--------|-------------------------------------------------------------------------------------------------------|
| TypeAugmented.md   | TaKE                  | oui    | ed Knowledge graph Embedding framework (TaKE) which could utilize type features to e                  |
| TypeAugmented.md   | TaRP                  | oui    | two views. Both TransT <sup>30</sup> and TaRP <sup>32</sup> collect relation types fro                |
| TypeAugmented.md   | TKRL                  | oui    | xploring the usage of type information. TKRL <sup>22</sup> extends traditional TransE                 |
| TypeAugmented.md   | TransC                | oui    | nct representations in different types. TransC <sup>23</sup> clearly distinguish concep               |
| TypeAugmented.md   | TransH                | oui    | of TransE have been developed. Such as, TransH <sup>13</sup> projects entity embeddings               |
| TypeAugmented.md   | TransR                | oui    | stead of relation-specific hyperplanes, TransR <sup>14</sup> directly projects entity e               |
| TypeAugmented.md   | TransT                | oui    | d jointly encodes these two views. Both TransT <sup>30</sup> and TaRP <sup>32</sup> coll              |
| TypeConstraints.md | Neural Tensor Network | oui    | deling of KGs. [20] recently proposed a neural tensor network, which we did not consider in our study |
| TypeConstraints.md | TransH                | oui    | network proposed in [20]. [23] proposed TransH which improves TransE's capability to m                |
| TypeConstraints.md | TransR                | oui    | further extended in [14] by introducing TransR which separates representations of enti                |
| Uniform.md         | Neural Tensor Model   | oui    | point. Another related approach is the Neural Tensor Model [14]. A special case of this model corr    |

## C1. Candidats faux négatifs - ÉVALUÉS (17)

| Article     | Entité | Où          | Extrait                                                                                |
|-------------|--------|-------------|----------------------------------------------------------------------------------------|
| ASA.md      | GCN    | prose+table | egating the neighbors' information. <b>R-GCN</b> (Schlichtkrull et al. 2018) considers |
| BatchNS.md  | GCN    | prose+table | al., 2018). The problem studied by the GCN is different than the one solved by PBG     |
| DHNS.md     | OTE    | prose+table | x [27], RotatE [25], PairRE [7], and GC-OTE [26]. (2) <b>Multimodal KGC models:</b> w  |
| DomainNS.md | KG2E   | table       | 0.2/td> </tr> <tr> <td>KG2E (He et al., 2015)</td> <td>No</td>                         |

| Article     | Entité       | Où          | Extrait                                                                                       |
|-------------|--------------|-------------|-----------------------------------------------------------------------------------------------|
| DomainNS.md | NTN          | table       | 6.7</td> </tr> <tr> <td>NTN (Socher et al., 2013)</td> <td>                                   |
| DomainNS.md | SE           | table       | </thead> <tbody> <tr> <td>SE (Bordes et al., 2011)</td> <td>                                  |
| DomainNS.md | Unstructured | table       | 9.8</td> </tr> <tr> <td>Unstructured (Bordes et al., 2014)</td> <td>                          |
| IGAN.md     | SE           | prose+table | results show that 1. Models including SE, SME(bilinear), TransE and TransH are f              |
| IGAN.md     | SME          | prose+table | ults show that 1. Models including SE, SME(bilinear), TransE and TransH are furthe            |
| IGAN.md     | Unstructured | prose+table | ages (Xie et al. 2016a) of entities. <b>Unstructured Model</b> (Bordes et al. 2012; Bordes et |
| LAS.md      | GCN          | prose+table | ls, e.g. ProjE [21], ANALOGY [22] and R-GCN [28], also achieve good results for lin           |
| LTS.md      | LFM          | table       | 1.3</td> </tr> <tr> <td>LFM</td> <td>469</td> <td>4                                           |
| LTS.md      | SME          | table       | 9.8</td> </tr> <tr> <td>SME</td> <td>542/526</td> <                                           |
| SelfAdv.md  | SE           | table       | </thead> <tbody> <tr> <td>SE (Bordes et al., 2011)</td> <td>\$-                               |
| STC.md      | TKRL         | prose+table | died Knowledge Representation Learning (TKRL) to take advantages of hierarchical ent          |
| Uniform.md  | SE           | prose+table | those of [3] (Structured Embeddings or SE) and [14]. 3 Table 1: **Numbers of pa               |
| Uniform.md  | SME          | prose+table | 2], and the energy-based models SE [3], SME(linear)/SME(bilinear) [2] and LFM [6].            |

## C2. Candidats faux négatifs - MENTIONNÉS (6)

| Article      | Entité    | Où    | Extrait                                                                                                |
|--------------|-----------|-------|--------------------------------------------------------------------------------------------------------|
| DMNS.md      | GraphSAGE | prose | conduct experiments using GAT [45] and GraphSAGE (SAGE) [14]. For GCN, we employ two lay               |
| GNDN.md      | GCN       | prose | sists of a Convolutional Graph Network (GCN) [39] based encoder for learning repres                    |
| HTENS.md     | TKRL      | prose | <sup>1</sup></td> </tr> <tr> <td>https://github.com/thunlp/TKRL SIGIR '23, July 23-27, 2023, Taipei, T |
| NSCaching.md | LINE      | prose | EGATIVE ENTITIES IN CACHE ON FB13. EACH LINE DISPLAYS 5 RANDOM SAMPLED NEGATIVE ENTI                   |
| RAAKGC.md    | KGT5      | prose | r encoding text descriptions. Moreover, KGT5-context (Kochsiek et al. 2023) and KG-S                   |

| Article            | Entité | Où    | Extrait                                                                                                                                      |
|--------------------|--------|-------|----------------------------------------------------------------------------------------------------------------------------------------------|
| TypeConstraints.md | SME    | prose | <sup>9</sup><br><a href="https://github.com/glorotxa/SME">https://github.com/glorotxa/SME</a><br>< <sup>10</sup> > Mainly caused by the rank |